

BACSWN SkyWatch Bahamas

The core technologies behind a sovereign, AI-operated meteorological watch, airspace surveillance, and emergency-response network for the Commonwealth of The Bahamas.

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BACSWN turns 15 island weather stations into a network of **intelligent, self-reliant agents** that sense, predict, warn, and coordinate the national response — and keep working through the strongest hurricanes, even when the central hub or an entire island is cut off. The technologies below are what make that possible.

15

INTELLIGENT STATIONS

3

INDEPENDENT COMMS LAYERS

40 TOPS

EDGE AI PER STATION

42

ALERT CHANNELS

CORE PLATFORM TECHNOLOGIES

Autonomous AI mesh network 15 weather stations that reason locally and reach consensus with their neighbors — continuing to sense, predict, and warn even if the hub or an island goes dark.	Triple-layer resilient communications Every station runs three independent links — a licensed VHF radio mesh, cellular, and hurricane-proof Iridium satellite — with automatic failover so no station is ever silenced.
Category 4/5-survivable hardware Towers, enclosures, and power engineered to ride through the strongest hurricanes (Dorian-class), with the satellite link as the guaranteed lifeline.	Edge artificial intelligence On-site NVIDIA Jetson processors run microclimate prediction and anomaly detection, continuously self-correcting forecasts against what actually happens.
End-to-end cryptographic security Hardware root-of-trust, national PKI, and encrypted, signed messaging on every link and device — FIPS-grade, with a post-quantum upgrade path.	Commercial + military airspace surveillance Tracks every aircraft in the Bahamas FIR by fusing ADS-B with Raytheon primary radar — covering military and non-cooperative traffic, not just airliners.

DATA, SENSING & RESPONSE TECHNOLOGIES

Tomorrow.io space-based weather radar Integrates commercial satellite weather radar to see storms forming over open ocean, where ground-based radar cannot reach.	Carbon-emissions tracking (CORSIA) Calculates per-flight CO ₂ to the ICAO CORSIA standard — turning the same overflight data into automated carbon accounting.
Emergency-services integration A shared Common Operating Picture and standards-based (CAP/EDXL) alerting connect NEMA, Police, Fire & Rescue, the Defence Force / Coast Guard, and the shelter system.	IoT sensor network Low-power LoRaWAN sensors densify coverage with street-level rain, flood, storm-surge, and lightning detection feeding the AI.
Autonomous drones (UAS) Drones for post-storm survey, search-and-rescue, and emergency comms relay — safely deconflicted using BACSWN's own live air picture.	International interoperability Built to exchange data with ICAO systems and, in a future phase, U.S. agencies — NOAA / National Hurricane Center, FAA Miami Oceanic, and NORAD.

HOW IT FITS TOGETHER

A single layer of **airspace surveillance feeds three missions at once**: aviation safety, carbon-emissions accounting, and overflight monitoring. The same mesh that carries weather warnings also drives **emergency coordination** across every Bahamian response agency. And because the survival core of every station is engineered to outlast a Category 5 storm, the network is at its most valuable precisely when the country needs it most.

Together these technologies make BACSWN a **sovereign, AI-operated safety network** — protecting Bahamian lives, airspace, and infrastructure, and continuing to operate when conventional systems fail.